

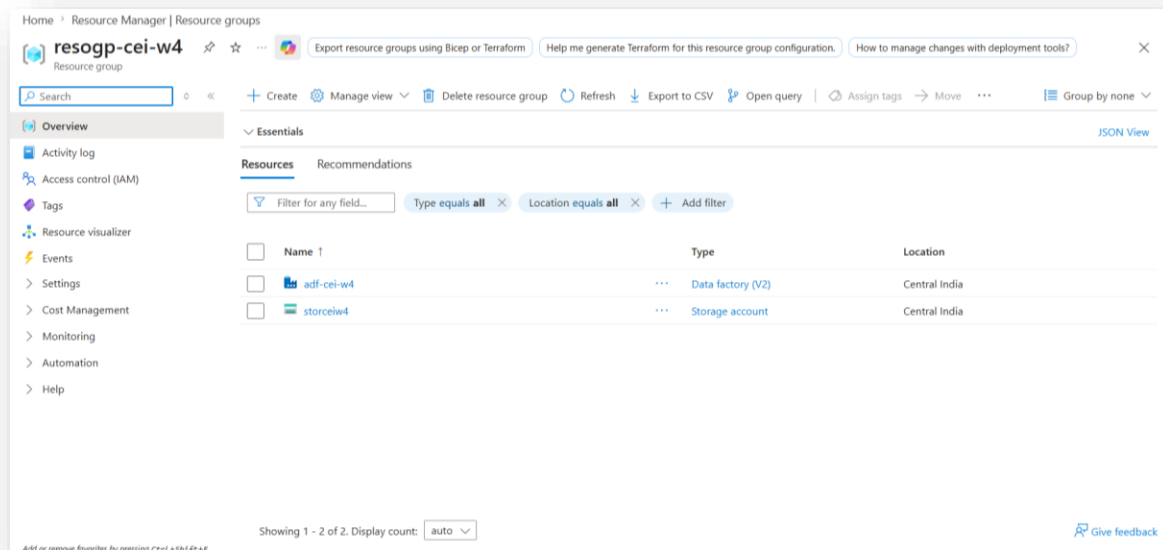
Week 4 Assignment

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

For this week's task, I focused on configuring the core Azure cloud components and then utilized them to build an end-to-end data pipeline in Azure Data Factory (ADF). The entire workflow was carried out using the Superstore sales CSV file as the primary dataset. The following sections include the screenshots for each activity, along with a clear explanation of the actions performed at every stage.

Task 1: Resource Group

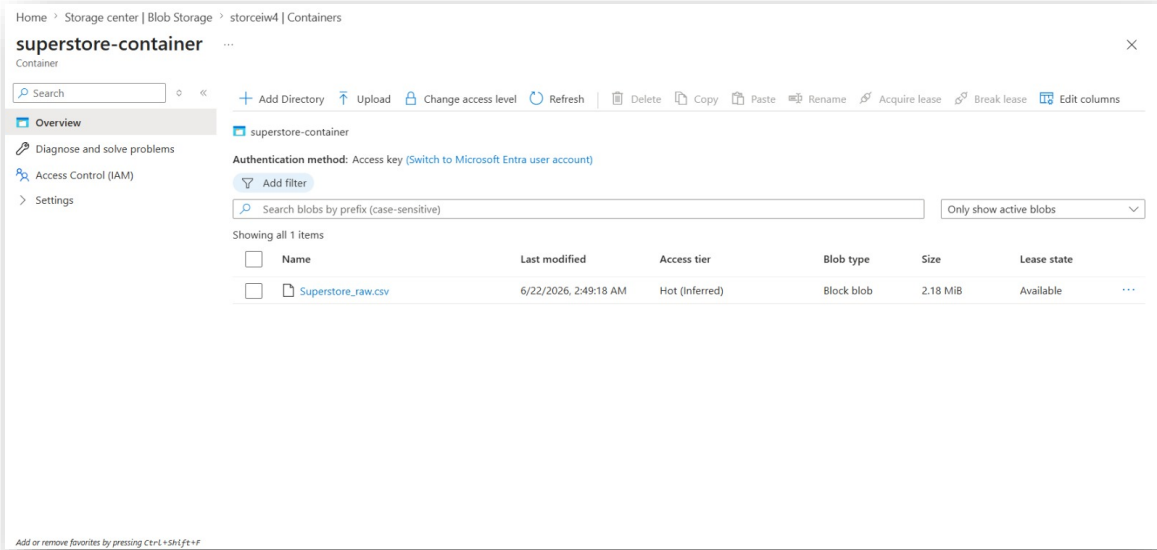
To begin the exercise, I set up a new resource group named **resogp-cei-w4** within the *Azure for Students* subscription, selecting the **Central India** region. This resource group serves as the central container for all the Azure components used in this week's work, including the storage account and the data factory instance.



Resource group resogp-cei-w4 showing the storage account and data factory inside it

Task 2: Storage Setup

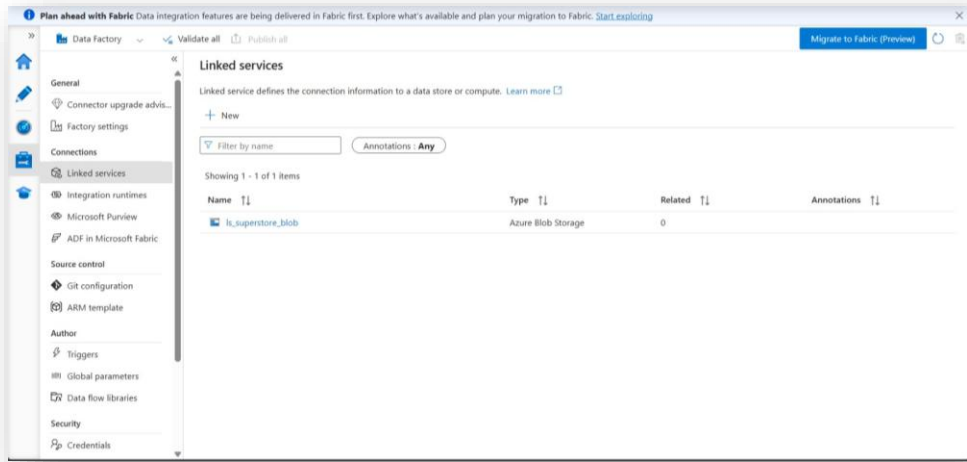
After that, I set up a new storage account named **storceiw4** and created a blob container inside it called **superstore-container**. I then uploaded the **Superstore_raw.csv** file into this container, which serves as the input dataset for the pipeline built later in the assignment.



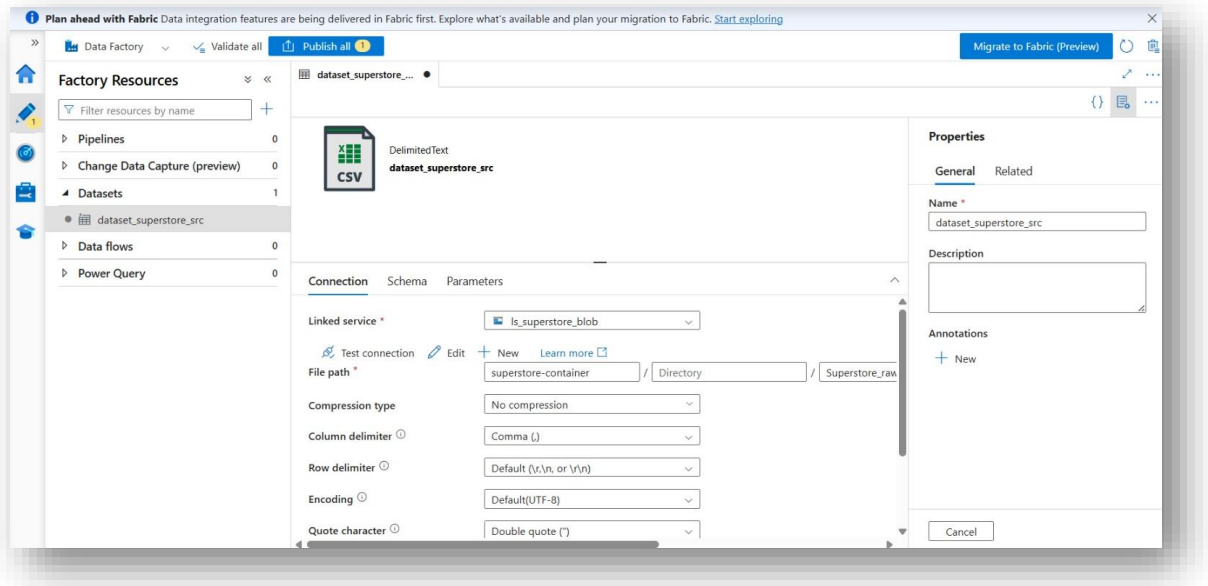
superstore-container showing Superstore_raw.csv uploaded (2.18 MiB)

Task 3: ADF Basics

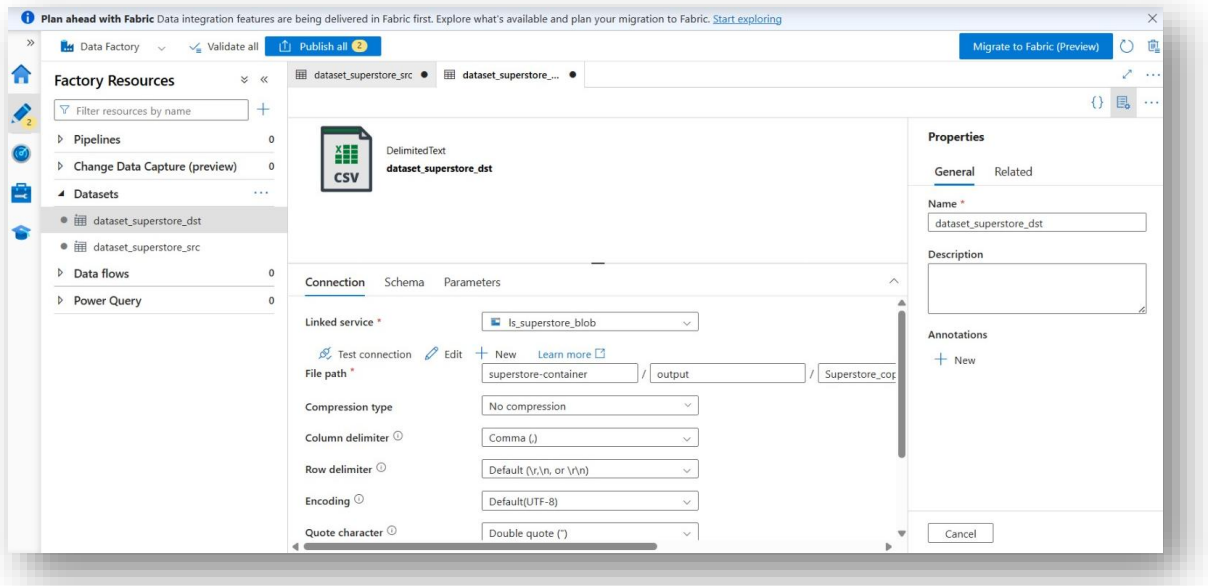
I then deployed an Azure Data Factory instance named **adf-cei-w4** and accessed ADF Studio to review the **Author**, **Monitor**, and **Manage** sections. Within the *Manage* area, I configured a Linked Service called **ls_superstore_blob**, which establishes the connection between ADF and the blob storage account. After setting this up, I created two datasets—one representing the source file and the other the output file.



Linked service ls_superstore_blob connected to Azure Blob Storage

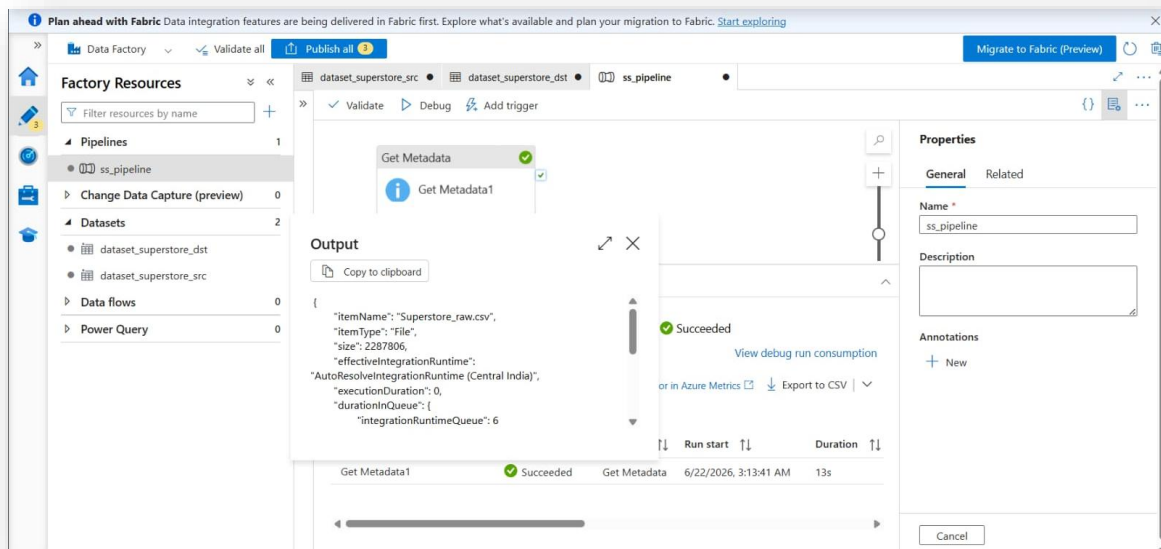


Source dataset `dataset_superstore_src` pointing to `Superstore_raw.csv`

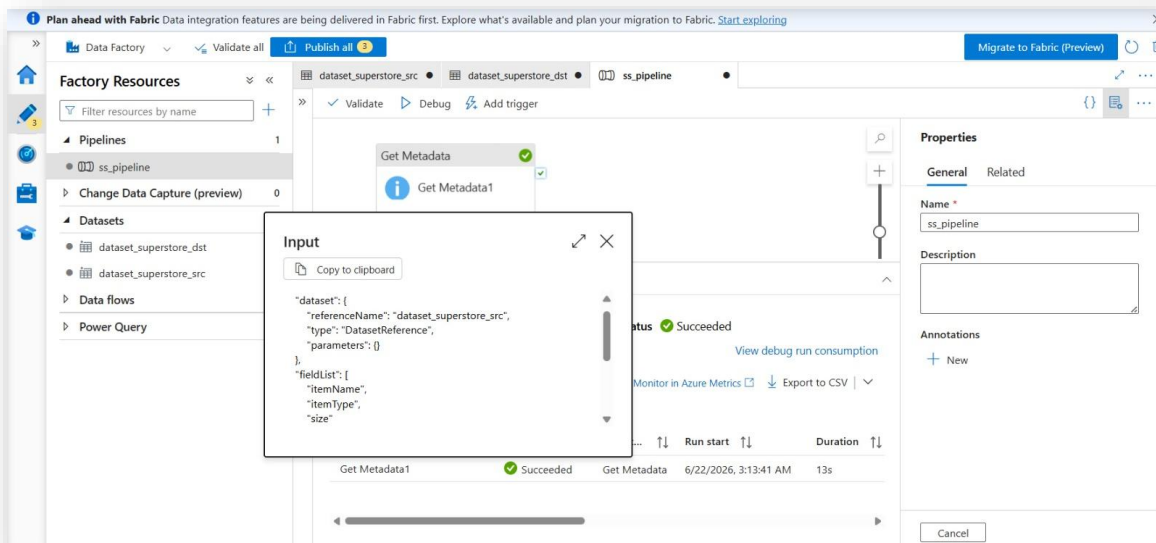


Destination dataset `dataset_superstore_dst` pointing to `output/Superstore_copy.csv`

After setting up the datasets, I included a **Get Metadata** activity in the pipeline to extract basic information from the source file—such as its name, format, and size. This step was mainly to verify that ADF could successfully access and interpret the file before proceeding with the actual copy operation.



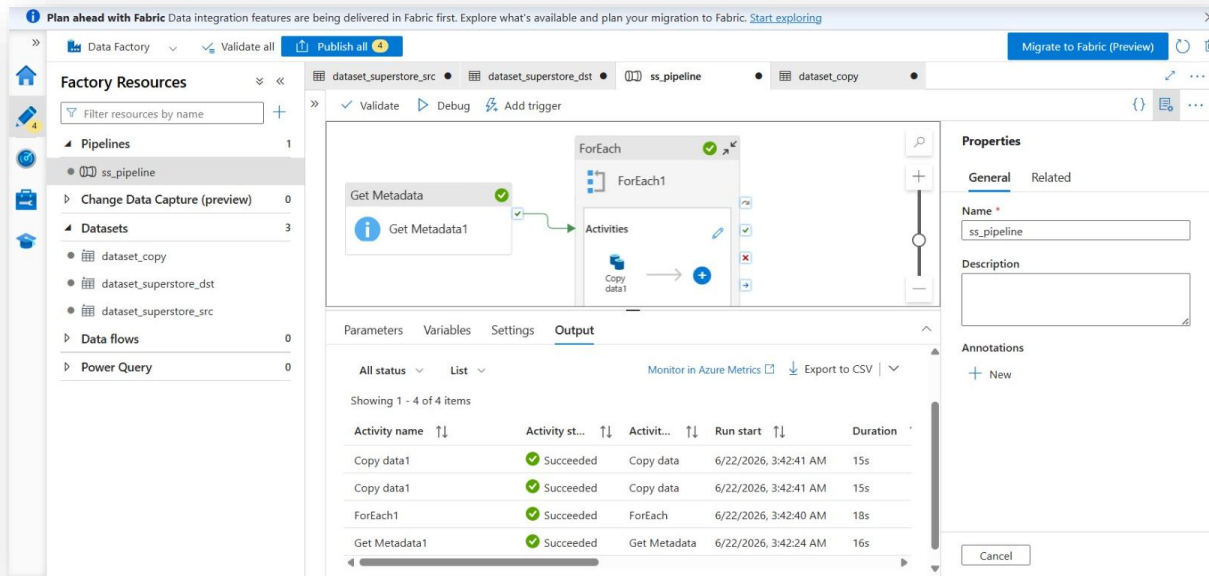
Get Metadata activity output showing file name, type and size of Superstore_raw.csv



Get Metadata activity input showing the dataset reference and field list used

Task 4: Pipeline Development

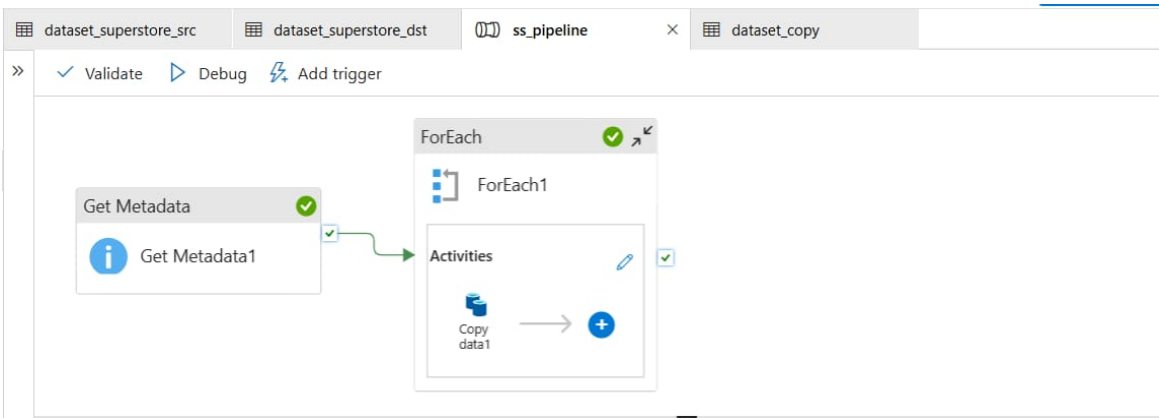
For the main pipeline, I configured a **Copy Data** activity to run after the **Get Metadata** step. Since the requirement included using a **ForEach** activity, I modified the metadata step to retrieve all child items within the container rather than targeting a single file. This list of items was then passed into the ForEach loop. Inside the loop, I placed the Copy Data activity so that each file detected in the container would be processed individually. I linked Get Metadata to ForEach using the success (green) dependency, ensuring that the copy operation only begins once the metadata extraction completes without errors.



Pipeline design: Get Metadata1 connected to ForEach1, with Copy data1 running inside the loop

Task 5: Pipeline Execution

I ran the pipeline using Debug mode. All four steps (Get Metadata1, ForEach1, and Copy data1 running twice inside the loop) completed with a Succeeded status.

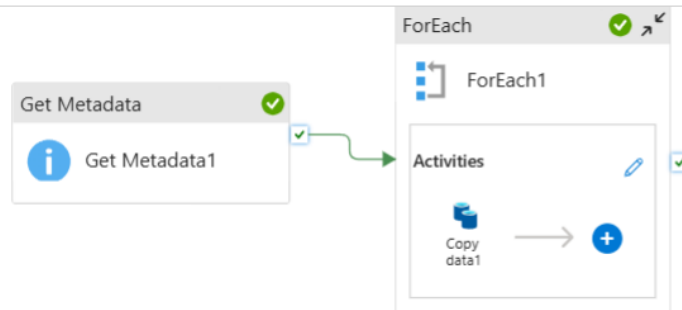


Pipeline canvas after a successful debug run — all activities show a green tick

Mini Project — Complete Pipeline (CSV Read + Process via ADF)

The mini project asked for a complete pipeline that reads a CSV file from Blob Storage and processes it using ADF, with a metadata check and the data copied to a new location. This is essentially the same pipeline built above (ss_pipeline), so I am using the same setup as proof for this part too:

- Source: Superstore_raw.csv in superstore-container (Blob Storage)
- Linked Service + Dataset + Pipeline: ls_superstore_blob, dataset_superstore_src/dst, ss_pipeline
- Process: Get Metadata check followed by Copy Data (inside ForEach)
- Destination: output/Superstore_copy.csv (new location in the same container)



Pipeline design used for the mini project

superstore-container > output

Authentication method: Access key ([Switch to Microsoft Entra user account](#))

Search blobs by prefix (case-sensitive) Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state	
☐	[.]						...
☐	Superstore_copy.csv	6/22/2026, 3:58:13 AM	Hot (Inferred)	Block blob	2.18 MiB	Available	...

Destination folder showing Superstore_copy.csv after the pipeline ran

In summary, the pipeline successfully accessed the CSV file from Blob Storage, verified its metadata through the **Get Metadata** activity, and transferred the data to the designated output location using **Copy Data**. This confirms that the pipeline executed as intended, the file was processed correctly, and all required validation steps were completed.